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Closure device and soft packaging container

Abstract

Provided is a closure device for closing a soft packaging container, the closure device including a holding section configured to hold a soft packaging container, a dispense section configured to dispense contents of the soft packaging container, a closure section configured to close the soft packaging container, and a control section configured to control timing for closing the soft packaging container by the closure section. The control section may include an acquisition section configured to acquire time information related to timing for closing the soft packaging container. Provided is a soft packaging container for accommodating contents, the soft packaging container including a main region for accommodating the contents, and an extension region provided by being extended from the main region.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
5213236	12/1992	Brown	N/A	N/A
6269837	12/2000	Arent	251/149.6	B67D 7/0205
6390329	12/2001	Maddox	222/25	A47K 5/1208
8240508	12/2011	Wegelin	222/105	G05B 15/02
8550378	12/2012	Mazooji	222/88	B05B 3/02
9504360	12/2015	Van Diepen	N/A	A47K 5/1217
9884336	12/2017	Hoefte	N/A	B05B 11/0054
10500699	12/2018	Chang	N/A	N/A
2005/0284888	12/2004	Rhodenbaugh	N/A	N/A
2009/0212072	12/2008	Fenton	222/173	B05B 11/0044
2009/0314783	12/2008	Losinski	N/A	N/A
2011/0114669	12/2010	Yang	222/639	A47K 5/16
2014/0103066	12/2013	Geert Norbert	N/A	N/A
2016/0101430	12/2015	Hoefte	N/A	N/A
2017/0043870	12/2016	Wu	N/A	N/A
2019/0210726	12/2018	Wu	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
104670706	12/2014	CN	N/A
105882973	12/2015	CN	N/A
107826251	12/2017	CN	N/A
110816842	12/2019	CN	N/A
2332453	12/2010	EP	N/A
H1084837	12/1997	JP	N/A

2010001007	12/2009	JP	N/A
2013224167	12/2012	JP	N/A
2017531492	12/2016	JP	N/A
2018516197	12/2017	JP	N/A
2019172355	12/2018	JP	N/A
20150003981	12/2014	KR	N/A
208691	12/1992	TW	N/A
200415081	12/2003	TW	N/A

OTHER PUBLICATIONS

ISA/237) Written Opinion of the International Search Authority for International Patent Application No. PCT/JP2021/005777, issued/mailed by the Japan Patent Office on Apr. 13, 2021. cited by applicant
ISA/237) Written Opinion of the International Search Authority for International Patent Application No. PCT/JP2021/005773, issued/mailed by the Japan Patent Office on May 11, 2021. cited by applicant
Office Action issued for related Japanese Application No. 2020-034282, transmitted from the Japanese Patent Office on Sep. 26, 2023 (drafted on Sep. 13, 2023). cited by applicant
Office Action issued for related Chinese Application 202180007368.9, issued by The State Intellectual Property Office of People's Republic of China on Aug. 24, 2024. cited by applicant
Office Action issued for counterpart Taiwanese Application 110106843, transmitted from the Taiwan Intellectual Property Office on Sep. 30, 2024 (issued on Sep. 25, 2024). cited by applicant
Office Action issued for related U.S. Appl. No. 17/801,267, issued by the US Patent and Trademark Office on Mar. 18, 2025. cited by applicant
Office Action issued for counterpart Chinese Application 202180015853.0, issued by The State Intellectual Property Office of People's Republic of China on Jun. 4, 2025. cited by applicant

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS
(1) This application is a National Stage Entry of International Application NO. PCT/JP2021/005777 filed in WO on Feb. 16, 2021, which claims priority to Japanese Patent Application NO. 2020-034292 filed in JP on Feb. 28, 2020, the contents of each of which are hereby incorporated by reference in their entirety.
BACKGROUND
1. Technical Field
(2) The present invention relates to a closure device and a soft packaging container.
2. Related Art
(3) Up to now, a pouch for accommodating contents has been known (for example, see Patent Document 1). Patent Document 1: Japanese Patent Application Publication No. 2019-172355
Technical Problem
(4) Up to now, control on timing for opening and closing a pouch has been difficult in some cases.
GENERAL DISCLOSURE
(5) According to a first aspect of the present invention, there is provided a closure device for closing a soft packaging container, the closure device including: a holding section configured to hold a soft packaging container; a dispense section configured to dispense contents of the soft

packaging container; a closure section configured to close the soft packaging container; and a control section configured to control timing for closing the soft packaging container by the closure section.
(6) The control section may include an acquisition section configured to acquire time information related to timing for closing the soft packaging container. The control section may close the soft packaging container according to the time information acquired by the acquisition section.
(7) The control section may include a sensing section configured to sense a content amount of the soft packaging container.
(8) The closure section may close the soft packaging container when a content amount of the soft packaging container becomes lower than a predetermined amount.
(9) The closure section may have a thermal welding mechanism configured to close the soft packaging container by thermal welding.
(10) The closure section may have a closure member applying mechanism configured to apply a closure member to the soft packaging container to be closed.
(11) The closure section may have a folding mechanism configured to fold the soft packaging container.
(12) The closure section may have a restraint section applying mechanism configured to restraint a folded section applied by the folding mechanism.
(13) The closure section may have a pinching mechanism configured to pinch the soft packaging container to be closed.
(14) The closure device may have a failure avoiding section configured to avoid failure upon closure by moving the contents in a closure region of the soft packaging container.
(15) The failure avoiding section may have a squeezing mechanism configured to squeeze the soft packaging container. The squeezing mechanism may squeeze contents of the soft packaging container to move the contents in the closure region.
(16) The failure avoiding section may have an orientation change section configured to change an orientation of the soft packaging container. The orientation change section may move the contents from the closure region by a change in an orientation of the soft packaging container.
(17) The closure device may have a cutting section configured to unseal the closed soft packaging container by cutting.
(18) The closure device may have a closure detection section configured to detect closure of the soft packaging container.
(19) The closure device may have an accommodation section configured to accommodate the soft packaging container. The accommodation section may be opened after closure of the soft packaging container.
(20) The dispense section may include an ejection device configured to eject the contents of the soft packaging container.
(21) According to a second aspect of the present invention, there is provided a soft packaging container for accommodating contents, the soft packaging container including: a main region for accommodating the contents; and an extension region provided by being extended from the main region.
(22) The extension region may be provided along an edge of the soft packaging container. The main region and the extension region may form a C-shaped structure.
(23) The extension region may have a length that is 50% or more of a width of the soft packaging container.
(24) The summary clause does not necessarily describe all necessary features of the embodiments of the present invention. The present invention may be a sub-combination of the features described above.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1**A illustrates an example of a configuration of a closure device **100**.
(2) FIG. **1**B illustrates an example of an embodiment of the closure device **100**.
(3) FIG. **2** illustrates an outline of the configuration of the closure device **100**.
(4) FIG. **3** illustrates an example of the configuration of the closure device **100** including a sensing section **44**.
(5) FIG. **4**A illustrates an example of a configuration of a closure section **30**.
(6) FIG. **4**B illustrates an example of the configuration of the closure section **30**.
(7) FIG. **4**C illustrates an example of the configuration of the closure section **30**.
(8) FIG. **5**A illustrates an example of a restraint section **36**.
(9) FIG. **5**B illustrates an example of the restraint section **36**.
(10) FIG. **5**C illustrates an example of the restraint section **36**.
(11) FIG. **6** illustrates an example of the closure section **30** having a pinching mechanism **37**.
(12) FIG. **7**A illustrates an example of a configuration of a failure avoiding section **60**.
(13) FIG. **7**B illustrates an example of the configuration of the failure avoiding section **60**.
(14) FIG. **7**C illustrates an example of the configuration of the failure avoiding section **60**.
(15) FIG. **8** illustrates an example of a cutting section **70**.
(16) FIG. **9**A illustrates an example of the closure device **100** including a closure detection section **80**.
(17) FIG. **9**B illustrates an example of the closure device **100** including the closure detection section **80**.
(18) FIG. **10** illustrates an example of the configuration of the closure device **100** including a locking mechanism.
(19) FIG. **11** illustrates an example of the configuration of the closure device **100**.
(20) FIG. **12** illustrates an example of a shape of a soft packaging container **150**.
DESCRIPTION OF EXEMPLARY EMBODIMENTS
(21) Hereinafter, the present invention will be described through embodiments of the invention, but the following embodiments do not limit the invention according to the claims. In addition, not all combinations of the features described in the embodiments necessarily have to be essential to solving means of the invention.
(22) FIG. **1**A illustrates an example of a configuration of a closure device **100**. The closure device **100** includes an accommodation section **110** and a lid section **120**. In addition, the closure device **100** includes a holding section **10**, a dispense section **20**, a closure section **30**, and a control section **40**. The present example illustrates states before and after the closure device **100** accommodates a soft packaging container **150**.
(23) The closure device **100** accommodates the soft packaging container **150** and dispenses contents **155** from the unsealed soft packaging container **150**. The closure device **100** closes the unsealed soft packaging container **150**. The closure device **100** accommodates the soft packaging container **150** in the accommodation section **110**. The lid section **120** is a lid of the accommodation section **110** accommodating the soft packaging container **150**. In the closure device **100**, the soft packaging container **150** can be taken out and put in by opening and closing the lid section **120**.
(24) The holding section **10** is configured to hold the soft packaging container **150**. For example, the holding section **10** has a hook shape to be hooked at an opening provided in the soft packaging container **150**. The holding section **10** may hold the soft packaging container **150** by nipping. The holding section **10** of the present example holds only an upper end of the soft packaging container **150** but may hold an entirety of the soft packaging container **150**.
(25) The dispense section **20** is configured to dispense the contents **155** from the soft packaging

container **150**. The dispense section **20** is inserted into the opening of the soft packaging container **150**. For example, the dispense section **20** ejects the contents **155** from an external nozzle of the closure device **100**.
(26) The closure section **30** is configured to close the unsealed soft packaging container **150**. Closure refers to closing of an outlet in a state where the contents **155** cannot be dispensed from the soft packaging container **150**. That is, the closure section **30** may change the soft packaging container **150** from an unsealed state into a sealed state, or may change the state into a state where the contents **155** cannot be dispensed by folding the opening of the soft packaging container **150** or the like.
(27) The control section **40** is configured to control timing for closing the soft packaging container **150** by the closure section **30**. For example, the control section **40** controls closure timing of the soft packaging container **150** according to a content amount, expiry date, or the like of the contents **155**. In an example, when an amount of the contents **155** of the soft packaging container **150** becomes lower than a predetermined amount, the control section **40** causes the closure section **30** to close the soft packaging container **150**.
(28) The soft packaging container **150** accommodates the contents **155**. The soft packaging container **150** includes a main region **151** and an extension region **152**. A capacity of the main region **151** is larger than a capacity of the extension region **152**. The extension region **152** is provided by being extended from the main region **151**. The extension region **152** is connected to the dispense section **20**. An opening for dispensing the contents **155** is provided in the extension region **152**, and is closed by the closure section **30**. The soft packaging container **150** will be described below.
(29) FIG. **1**B illustrates an example of an embodiment of the closure device **100**. The closure device **100** of the present example applies Japanese horseradish corresponding to the contents **155** to vinegared rice corresponding to an object **200**. The closure device **100** of the present example includes a nozzle **52**.
(30) For example, the closure section **30** closes the soft packaging container **150** after the end of business hours. In addition, the closure section **30** closes the soft packaging container **150** after the use of the contents **155**. In this manner, contamination of the closure device **100** or the like due to leakage of the contents **155** is prevented upon replacement of the soft packaging container **150**.
(31) The nozzle **52** is configured to extract the contents **155** towards the object **200**. The nozzle **52** may have a shape according to a use. The nozzle **52** of the present example has a shape appropriate to extract Japanese horseradish corresponding to the contents **155** to vinegared rice corresponding to the object **200**.
(32) FIG. **2** illustrates an outline of the configuration of the closure device **100**. The control section **40** includes an acquisition section **41** and a closure instruction section **45**.
(33) The acquisition section **41** is configured to acquire information required to decide the closure timing. The acquisition section **41** has a time information output section **42**, an information output section **43**, and a sensing section **44**.
(34) The time information output section **42** is configured to output time information related to timing for closing the soft packaging container **150**. For example, the time information output section **42** has a timer, and outputs elapsed time since the soft packaging container **150** is unsealed to the closure instruction section **45**. In addition, the time information output section **42** may have clock time acquisition means such as a clock, and output a current time to the closure instruction section **45**.
(35) The information output section **43** is configured to acquire information required to decide the closure timing of the soft packaging container **150**. For example, the information output section **43** outputs an expiry date of the soft packaging container **150** to the closure instruction section **45**. The information output section **43** may directly acquire the expiry date from the soft packaging container **150** accommodated in the closure device **100** or may acquire information input by a user

or the like.

(36) The sensing section **44** is configured to sense a content amount of the soft packaging container **150**. The sensing section **44** may sense a total capacity of the soft packaging container **150** before use or may sense a remaining amount of the contents **155**. For example, the sensing section **44** senses the content amount of the soft packaging container **150** based on a weight of the soft packaging container **150**.

(37) The closure instruction section **45** is configured to decide whether the soft packaging container **150** is to be closed according to input information. For example, the closure instruction section **45** closes the soft packaging container **150** according to at least one of the time information, the expiry date, or the content amount acquired by the acquisition section **41**. When the soft packaging container **150** is to be closed, the closure instruction section **45** inputs a closure command to the closure section **30**.

(38) FIG. **3** illustrates an example of the configuration of the closure device **100** including the sensing section **44**. The closure device **100** of the present example is different from the closure device **100** of FIG. **1**A in that the sensing section **44** is included. In the present example, an aspect different from the case of FIG. **1**A will be described in particular.

(39) The sensing section **44** senses a remaining amount of the contents **155** of the soft packaging container **150**. The sensing section **44** of the present example is a weight sensor configured to sense a remaining amount of the contents **155** by measuring a weight of the soft packaging container **150**. The sensing section **44** is coupled to the holding section **10** and measures the weight of the soft packaging container **150**.

(40) The control section **40** may take into account the weight of the soft packaging container **150** itself in the determination of the closure timing or does not need to take the weight into account. For example, the control section **40** closes the soft packaging container **150** when the sensed weight becomes lower than a predetermined value without taking into account the weight of the soft packaging container **150** itself. In addition, the control section **40** may acquire information from a two-dimensional code of the soft packaging container **150** and detect the remaining amount of the contents **155** by taking into account the weight of the soft packaging container **150** itself.

(41) FIG. **4**A illustrates an example of a configuration of the closure section **30**. The closure section **30** of the present example has a thermal welding mechanism **31** and a drive section **38**.

(42) The thermal welding mechanism **31** is configured to close the soft packaging container **150** by thermal welding. For example, the thermal welding mechanism **31** is a heat sealer for closing the soft packaging container **150**. The thermal welding mechanism **31** of the present example is arranged so as to face a closure position of the extension region **152**. In order to prevent thermal welding failure, the thermal welding mechanism **31** may be arranged in a region where it is difficult for the contents **155** to stay. The closure section **30** of the present example can prevent leakage of the contents **155** by closing an opening of the extension region **152** by the thermal welding mechanism **31**.

(43) The drive section **38** is configured to drive the thermal welding mechanism **31** so as to nip the extension region **152**. The drive section **38** of the present example nips the extension region **152** by moving a position of the thermal welding mechanism **31** by an air cylinder. The drive section **38** drives the thermal welding mechanism **31** at timing at which the control section **40** instructs closure of the soft packaging container **150**.

(44) The closure section **30** of the present example closes the extension region **152** in a state where the soft packaging container **150** is removed from the dispense section **20**, but may close the extension region **152** in a state where the dispense section **20** is inserted into the extension region **152**. In this case, the thermal welding mechanism **31** is arranged so as to avoid the dispense section **20**, and performs thermal welding of the extension region **152**.

(45) FIG. **4**B illustrates an example of the configuration of the closure section **30**. The closure section **30** of the present example has a closure member applying mechanism **32** and a closure

member **33**.
(46) The closure member applying mechanism **32** applies the closure member **33** to the soft packaging container **150** to be closed. For example, the closure member applying mechanism **32** nips the extension region **152** by the closure member **33** to close the soft packaging container **150**. For example, the closure member applying mechanism **32** is a robot hand that can move in a state where the closure member **33** is grabbed at will or release the closure member **33**.
(47) The closure member **33** is applied to the soft packaging container **150** to close the extension region **152**. The closure member **33** of the present example is a clip but is not limited to this. The closure member **33** may be a rubber band, an adhesive tape, or the like for closing the extension region **152**.
(48) The closure member applying mechanism **32** of the present example nips the extension region **152** with the clip by vertically moving the closure member **33**. After the closure member **33** is attached to the extension region **152**, the closure member applying mechanism **32** releases the closure member **33**. Therefore, the soft packaging container **150** can be removed to the outside of the closure device **100** in a state where the closure member **33** is mounted.
(49) FIG. **4**C illustrates an example of the configuration of the closure section **30**. The closure section **30** of the present example has a folding mechanism **34**.
(50) The folding mechanism **34** is configured to fold the soft packaging container **150**. The closure section **30** of the present example has two bar-shaped folding mechanism **34***a* and folding mechanism **34***b*. For example, the folding mechanism **34***a* and the folding mechanism **34***b* are robot fingers that can be manipulated at will.
(51) A restraint section applying mechanism **35** is configured to apply a restraint section **36** which will be described below to the extension region **152** so as to prevent the folded extension region **152** from returning to its original state. For example, the restraint section applying mechanism **35** closes the extension region **152** by thermal welding.
(52) In step S**100**, the folding mechanism **34***a* and the folding mechanism **34***b* nip the extension region **152**. Before step S**100**, the dispense section **20** may be extracted from the soft packaging container **150**, or after step S**100**, the dispense section **20** may be extracted from the soft packaging container **150**.
(53) In step S**102**, the extension region **152** is turned down by the folding mechanism **34***b* with the folding mechanism **34***a* set as a supporting point. In step S**104**, the extension region **152** is maintained in a folded state by the folding mechanism **34***b*.
(54) In step S**106**, while the folding of the extension region **152** is maintained by the folding mechanism **34***b*, the folding mechanism **34***a* set as the supporting point is extracted. In step S**108**, the restraint section applying mechanism **35** is pressed at a closure position of the extension region **152**. In step S**110**, the folding mechanism **34***b* is extracted. In this manner, the folding mechanism **34** realizes the folding and the thermal welding of the extension region **152**.
(55) FIG. **5**A illustrates an example of the restraint section **36**. The restraint section **36** is applied to the folded extension region **152** by the restraint section applying mechanism **35**. The restraint section **36** is configured to restraint a folded section applied by the folding mechanism **34**. Since the restraint section **36** of the present example is a clip, the closed soft packaging container **150** is easily unsealed.
(56) FIG. **5**B illustrates an example of the restraint section **36**. The restraint section **36** is a region where the extension region **152** has been subjected to thermal welding by the restraint section applying mechanism **35**. The extension region **152** of the present example is closed by folding, and maintains the folded state by thermal welding. That is, the restraint section **36** of the present example is obtained by thermal welding of mutual external surfaces of the extension region **152**, and it does not need to perform thermal welding of an inner side of the extension region **152** through which the contents **155** passes. Therefore, thermal welding failure hardly occurs due to residue of the contents **155**.

(57) FIG. **5**C illustrates an example of the restraint section **36**. The restraint section **36** is an adhesive tape. Since the restraint section **36** of the present example is an adhesive tape, the closed soft packaging container **150** is easily unsealed. In addition, the restraint section **36** of the present example can be disposed together with the spent soft packaging container **150**. Note that the restraint sections **36** illustrated in FIG. **5**A to FIG. **5**C may be mutually combined.
(58) FIG. **6** illustrates an example of the closure section **30** having a pinching mechanism **37**. The pinching mechanism **37** is configured to pinch the soft packaging container **150** to be closed. The closure section **30** of the present example has a pair of a pinching mechanism **37***a* and a pinching mechanism **37***b*. The pinching mechanism **37***a* has a mechanism configured to rotate about the pinching mechanism **37***b*. The closure section **30** pinches the extension region **152** by arranging the extension region **152** between the pinching mechanism **37***a* and the pinching mechanism **37***b* and rotating the pinching mechanism **37***a* for closing.
(59) The pinching mechanism **37** may be fixed to any configuration of the closure device **100**. For this reason, a configuration may be adopted where the soft packaging container **150** cannot be removed from the closure device **100** in a closed state by the pinching mechanism **37**. When the soft packaging container **150** is removed from the closure device **100**, the soft packaging container **150** may be separated from the pinching mechanism **37**. Therefore, in the closure device **100** of the present example, the pinching mechanism **37** can be repeatedly used even when the soft packaging container **150** is replaced.
(60) FIG. **7**A illustrates an example of a configuration of a failure avoiding section **60**. The failure avoiding section **60** of the present example has a squeezing mechanism **61**. The failure avoiding section **60** of the present example has two squeezing mechanism **61***a* and squeezing mechanism **61***b*.
(61) The failure avoiding section **60** is configured to prevent residue of the contents **155** by moving the contents **155** in the closure region of the soft packaging container **150**. In this manner, the failure avoiding section **60** avoids the failure upon closure of the soft packaging container **150**.
(62) The squeezing mechanism **61** is used to squeeze the soft packaging container **150**. The squeezing mechanism **61** is configured to move the contents **155** in the closure region by squeezing the contents **155** of the soft packaging container **150**. For example, the squeezing mechanism **61** is robot fingers for squeezing.
(63) In step S**200**, the squeezing mechanism **61***a* and the squeezing mechanism **61***b* are vertically moved to be arranged at positions corresponding to the closure region of the extension region **152**. In step S**202**, in a state where the squeezing mechanism **61***b* is fixed, the squeezing mechanism **61***a* is slid to an opening side of the extension region **152**. The squeezing mechanism **61***b* may be pressed against the extension region **152** to restrict movement of the contents **155** from the main region **151** side to the opening side.
(64) In step S**204**, the contents **155** in the closure region is discharged by sliding of the squeezing mechanism **61***a*. In step S**206**, thermal welding is performed by pressing the thermal welding mechanism **31** against the extension region **152** between the squeezing mechanism **61***a* and the squeezing mechanism **61***b*. In the present example, since the contents **155** in the closure region is discharged, the failure upon closure by the contents **155** can be avoided.
(65) FIG. **7**B illustrates an example of the configuration of the failure avoiding section **60**. The failure avoiding section **60** of the present example has an orientation change section **62**.
(66) The orientation change section **62** is configured to change an orientation of the soft packaging container **150**. For example, the orientation change section **62** changes an orientation of the extension region **152**. The orientation change section **62** moves the contents **155** from the closure region by the change in the orientation of the soft packaging container **150**. The failure avoiding section **60** of the present example includes an orientation change section **62***a* and an orientation change section **62***b*.
(67) The orientation change section **62***a* is connected to the dispense section **20** and changes a

position of the dispense section **20**. The orientation change section **62***a* of the present example is configured to change a position of the opening of the extension region **152** by changing a position of the dispense section **20**. The orientation change section **62***a* may change the position of the dispense section **20** in not only the vertical direction but also a horizontal direction.
(68) The orientation change section **62***b* is connected to the closure section **30** and changes a position of the closure section **30**. The orientation change section **62***b* may change the position of the closure section **30** according to control of the orientation change section **62***a*. For example, the orientation change section **62***b* may change the position of the closure section **30** in not only the vertical direction but also the horizontal direction according to a position of the extension region **152**.
(69) The orientation change section **62***a* of the present example raises the position of the opening of the extension region **152** by extending upwards. In this manner, since the contents **155** in the closure region of the extension region **152** flows onto an opposite side of the opening of The extension region **152**, the failure upon closure of the soft packaging container **150** can be avoided. Note that the orientation change section **62***b* of the present example may extend upwards or does not need to extend.
(70) FIG. **7**C illustrates an example of the configuration of the failure avoiding section **60**. Similarly as in FIG. **7**B, the failure avoiding section **60** of the present example has the orientation change section **62**. In the present example, a change degree of the orientation of the soft packaging container **150** is different from the embodiment of FIG. **7**B.
(71) The orientation change section **62***a* and the orientation change section **62***b* change the orientation of the soft packaging container **150** upside down by extending upwards. In this manner, since the contents **155** in the closure region of the extension region **152** flows to the main region **151**, the failure upon closure of the soft packaging container **150** can be avoided.
(72) The closure device **100** of the present example arranges the soft packaging container **150** outside the accommodation section **110** by extending the orientation change section **62**. For this reason, the soft packaging container **150** is closed, and the soft packaging container **150** can be immediately removed. Note that the closure device **100** may accommodate the soft packaging container **150** inside the accommodation section **110** even in a vertically inverted state.
(73) FIG. **8** illustrates an example of a cutting section **70**. The closure device **100** of the present example includes the cutting section **70** and a fixed section **72**. The extension region **152** has a thermal welding section **39** having been subjected to thermal welding by the thermal welding mechanism **31**.
(74) The cutting section **70** is configured to unseal the soft packaging container **150** by cutting the closed soft packaging container **150**. The cutting section **70** of the present example unseals the soft packaging container **150** by cutting the extension region **152**. In this manner, after the soft packaging container **150** is accommodated in the closure device **100** while being still sealed, the closure device **100** can unseal the soft packaging container **150**. The cutting section **70** may unseal the soft packaging container **150** in a state where the orientation of the soft packaging container **150** is fixed such that the contents **155** is not to be leaked.
(75) The fixed section **72** nips and fixes the extension region **152** in order to cut the extension region **152**. The fixed section **72** of the present example closes a path of the contents **155** in order to prevent the leakage of the contents **155** when unsealed. It it noted however that the fixed section **72** does not need to close the path of the contents **155** in such an orientation that the contents **155** is not to be leaked.
(76) Note that the fixed section **72** may have a thermal welding function of the extension region **152**. In this case, the closure device **100** can repeat closure and unsealing of the soft packaging container **150**. Since the extension region **152** is shortened when the closure and the unsealing of the soft packaging container **150** are repeated, positions of the closure section **30** and the cutting section **70** may be adjusted according to a length of the extension region **152**.

(77) FIG. **9**A illustrates an example of the closure device **100** including a closure detection section **80**. The closure device **100** of the present example includes a flowmeter as the closure detection section **80** which is configured to measure a flow rate of the contents **155**.
(78) The closure detection section **80** is configured to detect closure of the soft packaging container **150** by measuring a flow rate of the contents **155**. For example, the closure detection section **80** determines that the soft packaging container **150** is closed when the flow rate of the contents **155** is lower than a predetermined flow rate. The closure detection section **80** of the present example is provided in the dispense section **20**.
(79) According to another embodiment, the closure detection section **80** may calculate the flow rate of the contents **155** from a change in the weight of the soft packaging container **150** instead of measuring the flow rate of the contents **155**. In this case, the closure detection section **80** determines that the soft packaging container **150** is closed when a change amount of the weight of the soft packaging container **150** is lower than a predetermined value.
(80) FIG. **9**B illustrates an example of the closure device **100** including the closure detection section **80**. The closure device **100** of the present example includes a manometer as the closure detection section **80** which is configured to measure a pressure inside the accommodation section **110**.
(81) The closure detection section **80** is configured to detect closure of the soft packaging container **150** by measuring the pressure inside the accommodation section **110**. For example, in a case where the closure of the soft packaging container **150** is satisfactory, when a pressure source **82** is actuated, a pressure is increased. On the other hand, in the case of closure failure, since the soft packaging container **150** collapses, a pressure increase speed is decreased.
(82) FIG. **10** illustrates an example of the configuration of the closure device **100** including a locking mechanism. The closure device **100** of the present example is configured to prevent unintended opening and closing of the accommodation section **110** by locking the lid section **120**. For example, the closure device **100** opens the accommodation section **110** after the soft packaging container **150** is closed. In this manner, the closure device **100** prevents contamination by the contents **155** upon opening and closing of the accommodation section **110** due to carelessness of a user or the like. In addition, the closure device **100** may protect the user from high heat by locking the lid section **120** upon thermal welding.
(83) FIG. **11** illustrates an example of the configuration of the closure device **100**. The closure device **100** of the present example includes an ejection device **50**.
(84) The ejection device **50** is configured to eject the contents **155** of the soft packaging container **150**. The ejection device **50** of the present example is coupled to the dispense section **20** and ejects the contents **155** dispensed by the dispense section **20** from the nozzle **52**. For example, the ejection device **50** ejects the contents **155** using a pump. The ejection device **50** may decide an amount of the contents **155** to be ejected each time, a flow rate of the contents **155**, and the like.
(85) FIG. **12** illustrates an example of a shape of the soft packaging container **150**. The soft packaging container **150** of the present example has the main region **151** and the extension region **152**.
(86) The main region **151** and the extension region **152** are separated from each other by a slit C. A slit stop section may be provided at an edge of the slit C so as not to generate a fissure when the slit C is formed. For example, the generation of the fissure can be prevented by providing an opening at the edge of the slit C in advance.
(87) The extension region **152** is provided along an edge of the soft packaging container **150**. In this manner, the extension region **152** can freely adjust the position according to the position of the dispense section **20**. In addition, folding of the extension region **152** is facilitated. Then, the position of the opening of the extension region **152** can be easily freely adjusted according to the orientation of the soft packaging container **150**.
(88) The main region **151** and the extension region **152** form a C-shaped structure. Since the

extension region **152** is provided, the soft packaging container **150** of the present example more easily avoids thermal welding failure by the thermal welding mechanism **31** as compared with a case where the extension region **152** is not provided.

(89) A spatial region V is a region that is not occupied by the contents **155** inside the soft packaging container **150**. A volume of the spatial region V is equal to or more than a volume of the extension region **152**. In other words, the volume of the contents **155** in a still sealed state may be smaller than a spatial volume of the main region **151**. In this manner, the dispense section **20** is easily mounted since the extension region **152** can be emptied by accommodating the contents **155** in only the main region **151**. In addition, when the soft packaging container **150** is to be closed immediately after being unsealed, since the extension region **152** is not yet filled, closure failure hardly occurs. Furthermore, the soft packaging container **150** has a margin of the spatial region V in a still sealed state, handling such as folding is facilitated.

(90) The extension region **152** has a length that is 50% or more of a width W of the soft packaging container **150**. The extension region **152** may have a length that is 75% or more, or may have a length that is 90% or more, of the width W of the soft packaging container **150**. In the present example, the length of the extension region **152** is the same as the width W of the soft packaging container **150**.

(91) The length of the extension region **152** may be longer, or may be shorter, than the width W of the soft packaging container **150**. The length of the extension region **152** may be decided by taking into account an insertion length of the dispense section **20** into the extension region **152**, a length required for closure by the closure section **30**, a flow path length from the main region **151** to the extension region **152**, or the like. For example, the length of the extension region **152** is at least 40 mm or more. In addition, the length of the extension region **152** is 40 mm or more and may be equal to or less than the width W of the soft packaging container **150**.

(92) While the embodiments of the present invention have been described, the technical scope of the present invention is not limited to the above described embodiments. It is apparent to persons skilled in the art that various alterations and improvements can be added to the above-described embodiments. It is also apparent from the scope of the claims that the embodiments added with such alterations or improvements can be included in the technical scope of the present invention.

(93) The operations, procedures, steps, and stages of each process performed by an apparatus, system, program, and method shown in the claims, embodiments, or diagrams can be performed in any order as long as the order is not indicated by “prior to,” “before,” or the like and as long as the output from a previous process is not used in a later process. Even if the process flow is described using phrases such as “first” or “next” in the claims, embodiments, or diagrams, it does not necessarily mean that the process must be performed in this order.

EXPLANATION OF REFERENCES

(94) **10**: holding section, **20**: dispense section, **30**: closure section, **31**: thermal welding mechanism, **32**: closure member applying mechanism, **33**: closure member, **34**: folding mechanism, **35**: restraint section applying mechanism, **36**: restraint section, **37**: pinching mechanism, **38**: drive section, **39**: thermal welding section, **40**: control section, **41**: acquisition section, **42**: time information output section, **43**: information output section, **44**: sensing section, **45**: closure instruction section, **50**: ejection device, **52**: nozzle, **60**: failure avoiding section, **61**: squeezing mechanism, **62**: orientation change section, **70**: cutting section, **72**: fixed section, **80**: closure detection section, **82**: pressure source, **100**: closure device, **110**: accommodation section, **120**: lid section, **150**: soft packaging container, **151**: main region, **152**: extension region, **155**: contents, **200**: object

Claims

1. A closure device for closing a soft packaging container, the closure device comprising: a holding section configured to hold a soft packaging container; a dispense section configured to dispense

contents of the soft packaging container; a closure section configured to close the soft packaging container; and a control section configured to control timing for closing the soft packaging container by the closure section, wherein the control section includes an acquisition section configured to acquire time information related to timing for closing the soft packaging container, and the control section is configured to close the soft packaging container according to the time information acquired by the acquisition section.

2. The closure device according to claim 1, wherein the control section includes a sensing section configured to sense a content amount of the soft packaging container.

3. The closure device according to claim 1, wherein the closure section is configured to close the soft packaging container when a content amount of the soft packaging container becomes lower than a predetermined amount.

4. The closure device according to claim 1, wherein the closure section has a closure member applying mechanism configured to apply a closure member to the soft packaging container to be closed.

5. The closure device according to claim 1, wherein the closure section has a folding mechanism configured to fold the soft packaging container.

6. The closure device according to claim 5, wherein the closure section has a restraint section applying mechanism configured to restraint a folded section applied by the folding mechanism.

7. The closure device according to claim 1, comprising: a failure avoiding section configured to avoid failure upon closure by moving contents in a closure region of the soft packaging container.

8. The closure device according to claim 7, wherein the failure avoiding section has a squeezing mechanism configured to squeeze the soft packaging container, and the squeezing mechanism is configured to squeeze contents of the soft packaging container to move the contents in the closure region.

9. The closure device according to claim 7, wherein the failure avoiding section has an orientation change section configured to change an orientation of the soft packaging container, and the orientation change section is configured to move the contents from the closure region by a change in an orientation of the soft packaging container.

10. The closure device according to claim 1, comprising: a cutting section configured to cut the closed soft packaging container to be unsealed.

11. The closure device according to claim 1, comprising: an accommodation section configured to accommodate the soft packaging container, wherein the accommodation section is opened after closure of the soft packaging container.

12. The closure device according to claim 1, wherein the dispense section includes an ejection device configured to eject the contents of the soft packaging container.

13. The closure device according to claim 1, wherein the control section includes a sensing section configured to sense a content amount of the soft packaging container.

14. A closure device for closing a soft packaging container, the closure device comprising: a holding section configured to hold a soft packaging container; a dispense section configured to dispense contents of the soft packaging container; a closure section configured to close the soft packaging container; and a control section configured to control timing for closing the soft packaging container by the closure section, wherein the closure section has a thermal welding mechanism configured to close the soft packaging container by thermal welding.

15. A closure device for closing a soft packaging container, the closure device comprising: a holding section configured to hold a soft packaging container; a dispense section configured to dispense contents of the soft packaging container; a closure section configured to close the soft packaging container; and a control section configured to control timing for closing the soft packaging container by the closure section, wherein the closure section has a pinching mechanism configured to pinch the soft packaging container to be closed.

16. A closure device for closing a soft packaging container, the closure device comprising: a

holding section configured to hold a soft packaging container; a dispense section configured to dispense contents of the soft packaging container; a closure section configured to close the soft packaging container; and a control section configured to control timing for closing the soft packaging container by the closure section; and a closure detection section configured to detect closure of the soft packaging container.